Hardware Setup & First Power-On

Everything in this kit is already built, flashed and configured. Nothing to image, install or flash — plug it together, power it up, connect a phone, and it runs.

WHAT IS IN THE KIT

WiFi passwords and the SSH login are on the card supplied with the kit

ITEM	WHAT IT IS	NOTES
Raspberry Pi 4 base station	Runs the whole system	USB-C power supply. The **MT7612U USB WiFi adapter** is already fitted — leave it in.
ESP32-S3 timer watch	The caller's stopwatch	Waveshare 1.75" round AMOLED. Internal battery, charges over USB-C.
Bluetooth speaker	Field audio — calls and cues	UE WONDERBOOM, already paired to the Pi.
Your own phone or laptop	The CD's console	Any modern browser. Nothing to install.

POWER ON, IN THIS ORDER

1. **Speaker first.**
   Power the WONDERBOOM on and wait for its ready tone. The Pi reconnects to a speaker that is already awake; one woken later needs a nudge from *Settings*.
2. **Pi next.**
   Plug in the USB-C supply and allow **45–60 s** to boot and raise both WiFi networks.
3. **Watch last.**
   Short press the top-left button (L). It finds the timer network by itself.

NEVER HOLD THE TOP-RIGHT BUTTON AT POWER-ON

Holding R at boot forces ESP32 download mode and the screen stays blank. Power cycle it without touching any button.

Connect your console

Over WiFi in the field	Join `F3K_OPS`, then open `http://192.168.20.1:8080`. Any address you type is redirected there, so `go` in the bar is enough.
Over ethernet on a desk	Patch the Pi into your router and open `http://f3kbase.local:8080`. Needed only for software updates — it is the Pi's one route to the internet.

You should land on a dark page headed **GLIDE BASE**, with the tabs *Comp Setup · Rounds · Run · Results · Leaderboard · GliderScore · Settings*.

THE THREE NETWORKS

kept separate on purpose — the timer network carries no internet traffic

NETWORK	INTERFACE	ADDRESS	WHO JOINS IT
`F3K_BASE`	wlan1 — USB adapter	`192.168.10.1`	**Timer watches only.** Do not join this with a phone.
`F3K_OPS`	wlan0 — onboard	`192.168.20.1`	CD, scorer and spectator phones or tablets.
`eth0`	wired	DHCP — `f3kbase.local`	Optional. Internet for software updates only.

SET THE AUDIO UP

In **Settings → Audio Output**, pick one and press **Test**:

Bluetooth	Click the paired WONDERBOOM to connect. The competition setup.
3.5 mm Stereo	A powered speaker in the Pi's jack. Works with no setup at all.
USB Audio	Only selectable with a USB sound device plugged in.

Lead fires every cue a set number of seconds early to cancel Bluetooth's playback delay, and is tuned for this speaker. On the 3.5 mm jack set it back to 0 — and only tune it against the speaker you will compete with.

CONFIRM IT IS ALL UP

- ☐ Web UI loads, nav bar present
- ☐ **Settings → Connected Timers** lists the watch with a firmware version
- ☐ The watch shows `T1` and green `BASE OK`
- ☐ **Settings → Audio** shows the speaker connected; **Test** is audible
- ☐ Battery percentage showing on the watch

Grey `BASE...` means still associating — give it 20 s. Grey that never clears means the USB adapter did not come up; reboot with it plugged in.

THE WATCH'S TWO BUTTONS

stopwatch orientation — both buttons at 12 o'clock

BUTTON	POSITION	WHAT IT DOES
L — power key	top-left	Secondary: scratch, step a setting, page the log, open Round Recall.
R — BOOT	top-right	Primary: start and stop the flight, confirm, navigate. Hold to confirm.

Holding L is a hardware power-off the firmware never sees, so a long press always means R.

Two things that look like faults

- **The screen goes black.** AMOLED burn-in protection — it blanks after two minutes idle and any button wakes it. A live round never blanks.
- **Start buttons dead at idle.** Correct: the base owns round starts while the watch is connected. They unlock in the last two seconds of prep.

REFERENCE

If something is wrong

Web UI will not load	Check you are on `F3K_OPS`, not home WiFi
A WiFi network missing	`systemctl status hostapd`; reseat the adapter
Watch will not connect	Power cycle the watch; the base needs nothing
No audio	Wake the speaker, re-click it in Settings, check volume

Admin & shutdown

SSH host	`pi@f3kbase.local` — login on the card
Health check	`curl http://localhost:8080/health`
Logs	`journalctl -u f3k-server -n 50`
Shut down	`sudo poweroff`, then wait for the green LED to stop

Pulling power will not lose flights — they are written as they are flown — but copy the results from **Settings → Data Backup** first.

PLEASE LEAVE THESE ALONE — THIS IS A WORKING DEV RIG ON LOAN

Do not re-flash the watch, reset the Pi's network config, run anything under `setup/`, or press **Settings → Update from GitHub**. Everything an evaluation needs is in the web UI; nothing requires SSH.